LAB 2 – SpoilerAlert Software Requirements Specification

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1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and structural requirements for the SpoilerAlert prototype. This document is intended for the development team responsible for implementing the SpoilerAlert system and serves as a technical reference for design decisions, component integration, and verification. It is not intended as a marketing document or general-audience product overview.

SpoilerAlert is a cross-platform mobile and web application that aggregates federal product recall data from the Food and Drug Administration (FDA), the United States Department of Agriculture (USDA), and the Consumer Product Safety Commission (CPSC), and delivers personalized recall alerts by matching that data against user-maintained product inventories. The purpose of this SRS is to specify the system's behavior, interfaces, constraints, and functionality with sufficient precision to support prototype development.

1.2 Scope

SpoilerAlert addresses a persistent gap between the issuance of product recalls by U.S. federal agencies and the timely awareness of those recalls by individual consumers. The system aggregates recall data from the FDA OpenFDA API, the USDA FSIS API, and the CPSC API, cross-references that data against user inventory records, and delivers notifications via mobile push alerts and email.

The prototype will support the following core capabilities: user account management and authentication; product inventory management via manual entry, barcode scanning, and receipt scanning; automated recall monitoring across FDA, USDA, and CPSC data sources; a recall-matching engine that identifies when recalled products are present in a user's inventory; push and email notification delivery; allergen-based alert filtering; and a personalized dashboard summarizing active recalls relevant to the user.

The prototype will not include SMS/text message notification delivery, integration with grocery loyalty programs for automatic purchase tracking, or a premium subscription tier with advanced analytics. These features are deferred to a future release.

1.3 Definitions, Acronyms, and Abbreviations

Allergen: A substance, typically a protein, that causes an allergic reaction in certain individuals. Common food allergens include peanuts, tree nuts, milk, eggs, wheat, soy, fish, and shellfish.

Application Programming Interface (API): A set of protocols and tools that allows different software applications to communicate with each other. SpoilerAlert uses government APIs to retrieve recall data from FDA, USDA, and CPSC databases.

Consumer Product Safety Commission (CPSC): A U.S. government agency responsible for protecting the public from unreasonable risks of injury or death associated with consumer products.

Contamination: The presence of harmful substances in food or consumer products that make them unsafe for consumption or use.

Food and Drug Administration (FDA): A U.S. government agency responsible for protecting public health by ensuring the safety of food products, pharmaceuticals, cosmetics, and medical devices.

Immunocompromised: A condition in which an individual's immune system is weakened, making them more susceptible to infections and foodborne illnesses.

Model-Flow-Control-Display (MFCD): A software architecture pattern separating an application into four components: Model (data structures), Flow (navigation logic), Control (user interaction handling), and Display (user interface presentation).

Optical Character Recognition (OCR): Software technology that recognizes and converts text from images or photographs into machine-readable format, used in SpoilerAlert's receipt scanning feature.

Product Recall: The action of removing or correcting products that violate laws administered by government agencies due to safety risks.

Push Notification: An automated message sent from an application to a user's mobile device or web browser, appearing even when the application is not actively open.

Receipt Scanning: A feature that uses OCR technology to photograph and automatically parse grocery receipts, extracting product information to populate a user's inventory.

REST API: An application programming interface conforming to Representational State Transfer architectural constraints, used by SpoilerAlert to communicate with government recall databases.

Software Requirements Specification (SRS): A document describing the intended purpose, functionality, and constraints of a software system, serving as a contract between developers and stakeholders.

United States Department of Agriculture (USDA): A U.S. government agency whose Food Safety and Inspection Service (FSIS) issues recalls for meat, poultry, and processed egg products.

Universal Product Code (UPC): A standardized barcode that encodes product identification information, enabling SpoilerAlert to match scanned products against recall databases.

1.4 References

Adkins, J. (2026). Lab 1 – SpoilerAlert product description (Version 1). Old Dominion University, CS411W.

Centers for Disease Control and Prevention. (n.d.). Listeria outbreak linked to prepared pasta meals. https://www.cdc.gov/listeria/outbreaks/chicken-fettuccine-alfredo-06-25/index.html

Food Safety and Inspection Service. (2024, February 28). Summary of recall and PHA cases in calendar year 2023. U.S. Department of Agriculture. https://www.fsis.usda.gov/food-safety/recalls-public-health-alerts/annual-recall-summaries/summary-recall-and-pha-cases-0

Hallman, W. (2025). Recall modernization initiatives with the consumer in mind. In 2025 Food Safety Summit. https://www.foodsafety.com/ext/resources/FSS_Event/Presentations/2025/S7---Combined-Slides.pdf

Institute of Electrical and Electronics Engineers. (1998). IEEE recommended practice for software requirements specifications (IEEE Std 830-1998). IEEE.

U.S. Department of Justice. (n.d.). Conagra subsidiary sentenced in connection with outbreak of salmonella poisoning related to peanut butter. https://www.justice.gov/archives/opa/pr/conagra-subsidiary-sentenced-connection-outbreak-salmonella-poisoning-related-peanut-butter

1.5 Overview

The remainder of this SRS is organized as follows. Section 2, Overall Description, provides a system-level view of SpoilerAlert including its architectural context, primary functions, user roles, constraints, and dependencies. Future sections (not included in this submission) will provide detailed functional specifications, external interface requirements, and system quality attributes in accordance with IEEE 830-1998.

2 Overall Description

2.1 Product Perspective

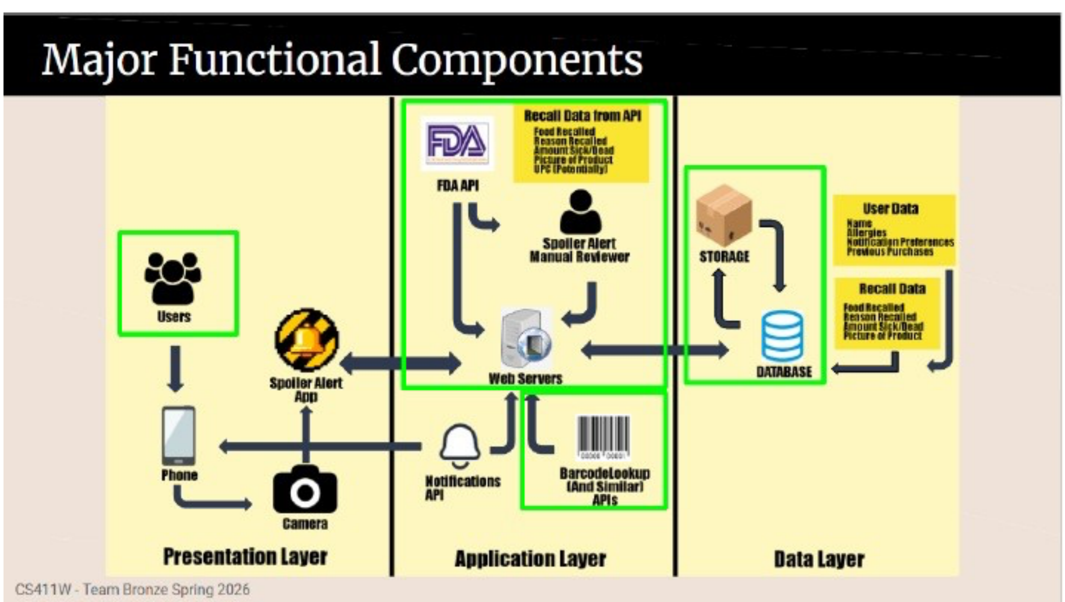
SpoilerAlert is a new, independent system that interfaces with existing federal agency recall data infrastructure to provide a consumer-facing recall notification service. It does not replace any government system; rather, it aggregates publicly available recall data through existing APIs and presents that data in a personalized, actionable format.

The system architecture follows a three-tier structure comprising a Presentation Layer, an Application Layer, and a Data Layer, consistent with the Major Functional Component Diagram (MFCD) established in Lab 1. The Presentation Layer encompasses the SpoilerAlert mobile application (Android and iOS) and web interface, through which users interact with inventory management, notification, and recall browsing features. The Application Layer hosts backend server processes responsible for recall data polling, the inventory-recall matching engine, notification dispatch, and barcode/receipt parsing services. The Data Layer consists of cloud-hosted database storage for user accounts, product inventory records, and cached recall data, with encryption applied to all user-sensitive information.

The system integrates with several third-party services via REST APIs. The FDA OpenFDA API provides food and pharmaceutical recall records. The USDA FSIS API supplies meat and poultry recall data. The CPSC API delivers consumer product safety recall information. Commercial barcode lookup services translate UPC codes into product metadata. Push notification services deliver real-time mobile alerts, and email service providers handle email notification routing. A manual data enrichment process supplements automated API data by linking UPC codes to recall records when government announcements lack specific product identification details.

Figure 1

Major Functional Components Diagram



2.2 Product Functions

The following table describes each planned feature of SpoilerAlert, along with its implementation status in the prototype. Features marked as Eliminated are deferred to a future release and will not be present in the prototype. Features marked as Partially Implemented are included with reduced scope or simulated data as noted.

Table 1

SpoilerAlert Feature Description and Prototype Implementation

Feature	Description	Prototype Implementation
User Account Management	Account creation, login, logout, password management	Fully Implemented
Barcode Scanning	Camera-based UPC scanning to add items to inventory	Fully Implemented
Receipt Scanning	OCR-based receipt parsing for batch inventory entry	Partially Implemented – OCR accuracy may vary; simulated fallback for parsing failures
Inventory Management	Add, edit, delete, and view product inventory entries	Fully Implemented
Recall Monitoring (FDA)	Automated polling of FDA OpenFDA API for food/pharmaceutical recalls	Fully Implemented
Recall Monitoring (USDA)	Automated polling of USDA FSIS API for meat/poultry recalls	Fully Implemented
Recall Monitoring (CPSC)	Automated polling of CPSC API for consumer product recalls	Partially Implemented – CPSC API coverage limited; manual enrichment used to supplement
Recall Matching Engine	Cross-reference user inventory against active recall records by UPC/product name	Fully Implemented
Push Notifications	Real-time mobile push alerts for recalls affecting user inventory	Fully Implemented
Email Notifications	Email delivery of recall alerts	Fully Implemented
Allergen Filtering	User-configurable allergen preferences for targeted alerts	Fully Implemented

Feature	Description	Prototype Implementation
Search & Browse Recalls	Search recalls by product name, brand, or UPC; browse active recall list	Fully Implemented
Dashboard	Personalized summary of active recalls affecting stored products	Fully Implemented
Manual Data Enrichment	Team member review linking UPC codes to recalls lacking product identifiers	Partially Implemented – applied selectively for high-impact recalls
Text Message Notifications	SMS delivery of recall alerts	Eliminated – deferred to future release
Grocery Loyalty Program Integration	Automatic purchase tracking via loyalty card APIs	Eliminated – deferred to future release
Premium Subscription Tier	Ad-free experience with advanced analytics	Eliminated – deferred to future release

2.3 User Characteristics

SpoilerAlert serves a single primary user role: the Consumer User. Consumer users are individuals who maintain personal product inventories and receive recall notifications through the system. No administrative or store-owner interfaces are present in the current prototype.

Consumer users are expected to have general smartphone proficiency and basic familiarity with mobile applications. No specialized technical knowledge is required. Users are assumed to have access to either an Android or iOS device, or a web browser on a desktop or laptop computer with internet connectivity.

The primary target population includes parents managing food allergens for children, immunocompromised individuals monitoring product safety, and health-conscious consumers seeking centralized recall awareness. These users are motivated by safety concerns rather than technical interest, and the system design prioritizes accessibility and clarity of information presentation accordingly. Users in this group are expected to interact with the application primarily through inventory scanning, notification review, and the recall dashboard.

A secondary user population consists of general consumers who conduct periodic recall lookups using the search and browse functionality without maintaining a persistent inventory. These users require minimal system expertise and interact with a subset of the system's features.

2.4 Constraints

The SpoilerAlert prototype operates under the following constraints.

API Availability: Recall data retrieval is dependent on the continued availability and schema stability of the FDA OpenFDA, USDA FSIS, and CPSC APIs. Changes to government API structures may require corresponding updates to the data ingestion layer.

Camera Hardware: Barcode scanning and receipt scanning features require access to a device camera. These features are unavailable on devices without camera hardware or where camera permissions are denied by the user.

Network Connectivity: Recall monitoring, notification delivery, and inventory synchronization require active internet connectivity. Offline functionality is limited to locally cached inventory data.

OCR Accuracy: Receipt scanning accuracy is subject to image quality, receipt formatting, and lighting conditions. The prototype may produce incomplete or incorrect results for receipts that do not conform to standard formatting.

2.5 Assumptions and Dependencies

The following assumptions and dependencies apply to the SpoilerAlert prototype.

Government API Access: The prototype assumes continued public availability of the FDA OpenFDA API, USDA FSIS API, and CPSC API at no cost. Any transition to authenticated or paid access would require architectural modification.

Barcode Lookup Service: The prototype depends on a third-party commercial barcode lookup service to translate UPC codes into human-readable product metadata. A service agreement with a barcode database provider is assumed.

Cloud Infrastructure: The prototype assumes access to a cloud hosting environment for backend server processes and database storage. Specific provider selection is a deployment-time decision and is not constrained by this specification.

Push Notification Service: The prototype depends on a third-party push notification provider compatible with both Android and iOS platforms.

Recall Data Completeness: The prototype assumes that government APIs provide sufficient product identification information to support automated recall matching for the majority of recall records. Manual enrichment is assumed to be necessary for a minority of records where product identification data is absent from the API response.